Alexandre Autran

Normalien in Machine Learning and Mathematics

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Personal Webpage | LinkedIn (+2k followers) | GitHub | Google Scholar

Presentation

I am a machine learning and mathematics student at the Écoles Normales Supérieures (ENS) in Paris-Saclay and Rennes. My research interests are centered on probabilistic and geometric methods for machine learning, high-dimensional statistics, density analysis, optimal transport, graph-based learning, and statistical learning theory.

Education

2024–2027 | **ENS Diploma, Department of Mathematics**
Admitted to the ENS through a competitive entrance examination.

2026–2027 | **École Normale Supérieure Paris-Saclay**, Paris, France
Research year in machine learning and applied mathematics.
Final year of the ENS Diploma (research track at ENS de Rennes).

2024–2027 | **École Normale Supérieure de Rennes**, Rennes, France
MSc in Mathematics (year 2). Highest honors (17/20), first semester.
MSc in Mathematics (year 1). Highest honors (17.6/20). Rank: 5th/60+.

2024–2025 | **École Normale Supérieure de Rennes**, Rennes, France
BSc in Mathematics (final year), Computer Science minor. High honors (15.4/20).

2022–2024 | **University of Rennes**, Rennes, France
BSc in Mathematics (years 1 and 2), Computer Science minor.
Selective Défi track. Highest honors (18.5/20 and 18/20). Rank: 1st-2nd/100+.

Research Experiences

Aug 2026– | **Machine Learning Researcher**
High-Dimensional Density Analysis Using Local Unimodality.
Supervised by Argyris Kalogeratos at Centre Borelli, ENS Paris-Saclay, France.

May–Aug 2026 | **MSc Research Internship**
Thermochromic Modeling of the Emissivity of Oxide Nanocomposites. [Report] [Presentation]
Supervised by Nguyen Ngoc Duy and Amaury Baret at the University of Liège, Belgium.

Apr–Jun 2026 | **MSc Research Internship**
Numerical Simulation and Optimal Control of Spatial Lotka–Volterra Systems. [Report]
Supervised by Martin J. Gander and Bastien Chaudet-Dumas at the University of Geneva, Switzerland.

Apr–May 2026 | **Machine Learning Projects**
Cardiovascular Disease Prediction Using Statistical Learning and Neural Network Models. [Report]
Air Pollution Prediction from Environmental Data. [Report]
Studies carried out as part of the MSc course *Statistical Learning* (prof. Myriam Maumy).

Jan 2026 | **Statistical Reading Group**
Supervised Statistical Learning: Prediction by Empirical Risk Minimization. [Presentation]
Supervised by Magalie Fromont at ENS de Rennes, France.

Nov–Dec 2025 | **MSc Research Seminar**
Point Process and Stein's Method. [Report] [Presentation]
Supervised by Jean-Christophe Breton at the University of Rennes, France.

Oct–Nov 2025 | **Algebra Reading Group**
Elimination Theory and Extension. [Presentation]
Supervised by Jade Nardi at ENS de Rennes, France.

Jun–Aug 2025 | **BSc Research Internship**
Lattice-Based Post-Quantum Cryptography. [Report] [Presentation]
Supervised by Pierre-Alain Fouque and Aurore Guillevic at Inria, France.

May–Jun 2025 | **BSc Research Internship**
Algebraic Counting of the Real Roots of Polynomials. [Report] [Presentation]
Supervised by Marie-Françoise Roy at the University of Rennes, France.

Feb–Mar 2025 | **Directed Research Reading**
Geometry of Numbers. [Report] [Presentation]
Supervised by Juliette Bavard at the ENS de Rennes, France.

Feb–Mar 2024 | **BSc Research Internship**
Topological Study of Matrix Groups. [Report]
Supervised by Françoise Dal'Bo and Jérémy Bettinger at the University of Rennes, France.

May–Jun 2023 | **BSc Research Internship**
Introduction to Hyperbolic Geometry and Fuchsian Groups. [Report] [Presentation]
Supervised by Frank Loray at the University of Rennes, France.

Publications

[4] **A. Autran**, "Mean-Field Fluctuations under Emergent α-Stable Common Noise", Preprint, 2026.

[3] **A. Autran**, "Second-Order Asymptotics of Emergent α-Stable Common Noise", Preprint, 2026.

[2] **A. Autran**, "Optimization with an Exact Discrete Adjoint for a Diffusive Lotka–Volterra Harvesting Problem", Preprint, 2026.

[1] **A. Autran**, D. Munteanu, and J.-L. Autran, "Multi-Dimensional Charge Diffusion–Collection Model in Semiconductor Devices Subjected to Single Ionizing Particles", *Applied Sciences*, vol. 16, no. 11, Art. no. 5551, 2026. [Paper]

Other Activities

2024–2025 | **ENS Oral Examinations**
Supervision of 2nd year BSc students in preparation for the oral entrance examinations in mathematics of the Écoles Normales Supérieures: Ulm, Paris-Saclay, Rennes and Lyon. [Resources]

2022–2025 | **Teaching in Mathematics**
Tutoring for final-year high school students of the Lycée Chateaubriand in Rennes, and for first and second years BSc students of the University of Rennes. [Resources]

2023–2024 | **BSc Student Representative**
Representative on the council of the Mathematics Department of the University of Rennes.

Before 2022 | **High School Competitions**
Concours Général in Mathematics, Physics, Philosophy, and National High School Mathematics Olympiad at Lycée Thiers in Marseille.

Computer Skills

Programming: Python, C/C++, Java, MATLAB, R. **Python / ML:** NumPy, pandas, scikit-learn, PyTorch.

Languages

French: native language. **English:** professional working proficiency.

Sports and Interests

Sports: running, cycling, strength training. **Interests:** reading and writing, in fantasy and science fiction literature; chess.

Academic Reference

François Bolley [Web page]
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